Zixiang (Zach) Tong

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SUMMARY

Ph.D. candidate in Engineering Mechanics with expertise in **digital image and volume correlation (DIC/DVC)**, **material characterization**, and **FE model validation**. Experience in thermo-mechanical FE modeling, inverse material-property identification, and development of validated measurement software. Available April 2027.

EDUCATION

Ph.D., Engineering Mechanics (expected April 2027) — The University of Texas at Austin, 2023 – present
Advisor: Dr. Jin Yang

M.S., Solid Mechanics — Southeast University, 2020 – 2023

B.Eng., Engineering Mechanics — Southeast University, 2016 – 2020

SKILLS

Solid Mechanics: Continuum and nonlinear mechanics, elasticity/plasticity, thermo-mechanical stress analysis, large deformation, contact, residual/interfacial stress, rate-dependent behavior, fracture

FEA & Simulation: Abaqus, COMSOL; implicit static, quasi-static and thermo-mechanical analysis; multi-material assemblies; PDE-constrained inversion and adjoint methods; parametric studies; solver automation

Material Characterization & V&V: Constitutive parameter and property-field identification, instrumented and **high-strain-rate** testing, full-field 3D strain measurement (stereo-DIC/DVC), simulation–experiment correlation, verification and validation

Machine Learning: PyTorch, GPU training, dense correlation and optical flow, **FE-based surrogate modeling**, synthetic training data, synthetic-to-experimental transfer, model benchmarking

Programming: Python, MATLAB, C/C++, Numba, Git, Linux; automated pre/post-processing, batch solver execution, packaged and tested software

EXPERIENCE

Heterogeneous Material Property Identification — UT Austin *2025 – present*

- Co-developed an **inverse identification framework** recovering spatially varying **constitutive properties** from a single instrumented test.
- Simultaneously identified material properties in each phase and interface geometry without prescribing the interface location.
- Applied **measured boundary displacements** to the FE model, reducing reliance on idealized boundary conditions in simulation–experiment correlation.
- In collaboration with the Oden Institute and Sandia National Laboratories.

Full-Field Strain Measurement and FE Model Validation (Stereo-DIC / DVC) *2020 – present*

- Developed and published a **3D stereo-DIC** framework for full-field displacement and strain measurement, providing FE **validation data** and measured boundary conditions.
- Implemented augmented-Lagrangian optimization, adaptive meshing, and mask-aware subset splitting for displacement measurement near holes and cracks, with strain estimation within valid material regions.
- Evaluated systematic bias and measurement noise using reference cases to assess the reliability of displacement and strain fields for model validation.
- Developed classical and learning-based measurement workflows spanning **2D DIC, stereo-DIC, and volumetric DVC**.
- Developed **RAFTcorr** in PyTorch, achieving **100–1,000× speedup** over iterative DIC in benchmark tests; characterized accuracy and operating limits.
- Developed **RAFT-DVC**, a resolution-aware volumetric solver with **100–1,000× speedup** over iterative DVC in benchmark tests; demonstrated transfer from synthetic training data to experimental micro-CT volumes.
- Design and build **DIC/DVC experimental setups**: optical layout, calibration, and synchronized high-speed acquisition, hardware-synchronizing **FLIR and PixeLINK camera arrays** and **Shimadzu HPV-X2** ultra-high-speed imaging through **NI** hardware, with **Arduino**-controlled motion stages and illumination triggered in lock-step with capture.

High-Strain-Rate Material Characterization (Laser-Induced Cavitation) *2024 – present*

- Use **laser-induced cavitation** imaged at **1 Mfps** (Shimadzu HPV-X2) to measure transient material response under rapid loading.
- Investigate **rate-dependent material response** and interfacial damage accumulation under repeated cavitation loading.
- Extended the approach to liquid-filled architected materials and developed automated interface tracking and batch analysis of high-speed image sequences.

Package-Level Thermo-Mechanical FEA and Test Correlation — UT Austin | SRC-sponsored *2025 – present*

- Built and ran **implicit thermo-mechanical FE models in Abaqus** of layered multi-material chip packages with temperature-dependent properties and CTE mismatch across bonded layers, predicting warpage and interfacial stress under thermal **load cases**.
- Ran parametric studies over geometry, layer thickness, materials, and thermal loading to rank the dominant contributors to warpage.
- Quantified the **simulation-to-test discrepancy** against full-field measurements and used the residuals to calibrate model inputs.

Advanced Packaging R&D Intern — Tokyo Electron (TEL) *Austin, TX | May 2026 – Aug. 2026*

- Developed **FE-based ML surrogate models** for advanced packaging applications.
- Built an end-to-end pipeline from parametric FE multiphysics simulations to trained models, covering data generation, training, and evaluation.
- Integrated simulation outputs and measurement data into a model evaluation and calibration workflow. *(Details confidential.)*

Computer Vision Engineer Intern — Orbbec Inc. *Shenzhen, China | Summer 2022*

- Built Python/C++ calibration and validation tooling for production 3D camera hardware; diagnosed system-level measurement error against production accuracy requirements.

ENGINEERING SOFTWARE (OPEN SOURCE)

github.com/zachtong

- **pyALDIC** — open-source augmented-Lagrangian DIC software with GUI, scriptable API, automated tests, and documented verification cases. Recommended in **two UT Austin undergraduate courses (100+ students)**; distributed through PyPI/GitHub/Zenodo under BSD-3, with stereo-surface and volumetric variants.
- **RAFTcorr** — deep-learning DIC software with a GUI analysis pipeline and documented accuracy, runtime, and operating limits.
- **RAFT-DVC** — learning-based volumetric displacement measurement, with resolution-aware benchmarking against tuned classical solvers.

SELECTED PUBLICATIONS

- A. Jones, S. DiCecco, . . . , **Zixiang Tong**, J. Yang. E. Jones and M. Iadicola, editors. A Good Practices Guide for Digital Image Correlation, 2nd ed. *International Digital Image Correlation Society (iDICs)*, 2025.
- J. Kirchhoff, **Zixiang Tong**, et al. Extracting Heterogeneous Material Properties using an Infinite-Dimensional Inversion Digital Image Correlation Framework. *In preparation*, 2026.
- **Zixiang Tong**, et al. 3D Stereo Adaptive Mesh Augmented Lagrangian Digital Image Correlation. *Experimental Mechanics*, 2025.
- **Zixiang Tong**[†], et al. Machine Learning-Aided Spatial Adaptation for Improved Digital Image Correlation Analysis of Complex Geometries. *Strain*, 2026.
- **Zixiang Tong**, et al. RAFTcorr: A Deep Learning Digital Image Correlation Framework with Operating-Boundary Characterization. *Preprint*, 2026. doi:10.21203/rs.3.rs-9452500
- **Zixiang Tong**, et al. pyALDIC: Augmented Lagrangian Digital Image Correlation in Python, with Adaptive Meshing and Mask-Aware Subset Splitting. *arXiv:2607.22755*, 2026.
- **Zixiang Tong**, et al. RAFT-DVC: Resolution-Aware Machine Learning-Based Digital Volume Correlation. *arXiv:2609.01876*, 2026.
- **Zixiang Tong**, et al. Refraction Error Analysis in Stereo Vision for System Parameters Optimization. *Measurement*, 222, 113650, 2023.

[†] Co-first author

CONFERENCE PRESENTATIONS

- International DIC Society Conference — Alexandria, VA *Nov. 2025*
- SRC TECHCON — Austin, TX *Sep. 2025*
- SEM Annual Conference — Milwaukee, WI *Jun. 2025*
- IMAC Conference — Orlando, FL *Feb. 2025*
- SEM Annual Conference — Vancouver, WA *Jun. 2024*

AWARDS

- **SRC Research Scholar**, Semiconductor Research Corp., 2025
- **Graduate Recognition Award**, UT Austin, 2025 | **Graduate Excellence Fellowship**, UT Austin, 2023–2026